

"Can I use my mouth to navigate in VR?": Integrating Mouth Gestures in XR with (and for) Disabled Artists through Criptastic Hacking

Puneet Jain
Individualized Studies
Concordia University
Montreal, Quebec, Canada
Immersive Arts Space
Zurich University of the Arts
Zurich, Switzerland
puneet.jain@mail.concordia.ca

Chris Lloyd Salter
Immersive Arts Space
Zurich University of the Arts
Zurich, Switzerland
christopher.salter@zhdk.ch

Abstract

Despite the rise of affordable XR technologies, accessibility still remains a key concern, often excluding people with disabilities from accessing immersive XR platforms. This work highlights these issues through a case study centered on an XR artwork: *Crip Sensorama*, co-created by two disabled artists and an HCI researcher. First, we present the accessibility challenges encountered by the artists to navigate and interact in XR. Secondly, we introduce the methodology of Criptastic Hacking, which led to the creation of two sets of mouth gestures for the artists as an interaction modality in XR. Finally, we share insights on how integrating this unconventional but accessible input modality into the artwork allowed us to incorporate the lived experiences of disability within HCI-based frameworks, that encourages inclusive XR design. Through this work, we aim to share new approaches to accessible XR design through inter-able collaborations, potentially reshaping future HCI research and practice.

CCS Concepts

• Human-centered computing → Accessibility technologies.

Keywords

eXtended Reality, Assistive Technology, Media Art, Disability Studies, Inclusive Design

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1 Introduction

Following the rise of affordable consumer-focused XR technologies (by companies like *Meta*), disabled artistic communities have started exploring these platforms for creative expression and immersive experiences [22] [26] [20]. XR (or eXtended Reality) is an umbrella term for Virtual/Augmented/Mixed Reality (VR/AR/MR) experienced using head-mounted displays and hand-held controllers. However, the reliance on head movements for 360-degree point of view and rapid hand movements to interact with these devices poses significant barriers to people with motor disabilities such as muscular dystrophy or spinal cord injuries. The inaccessibility of these XR devices thus exposes what Gerling and Spiel refer to as the 'corporeal standard', wherein the non-disabled are the default target users [5]. This standard thus obstructs disabled artists and activists from integrating XR devices into their artistic practices, limiting them to not share their lived experiences of disability and art-making through XR.

The discipline of Human-Computer Interaction has expanded its focus on creating accessible XR solutions for people with varied disabilities such as Blind and Low Vision [24] [11], Deaf or Hard of Hearing [13] [21] [1], Autism [17] and Invisible disabilities [6]. Although solutions have also been proposed for people with limited motor movements [3] [29] [16], or who use wheelchairs [4] [19], which allow them to interact in XR, there is little to no research for people who have almost no motor movements in the upper and lower parts of their bodies. While non-XR art-works such as EyeWriter [14] demonstrate that low-cost eye-tracking as a feature can be an effective tool for artistic expression for people with severe motor disabilities, only a few headsets such as Meta Quest Pro and Apple Vision Pro support this feature. In addition, these headsets remain prohibitively expensive for broader acceptability and are bulky, specifically for our collaborators and disabled artists Eric Desrosiers (quadriplegic) and Christian Bayerlein (tetraplegic) - who favor alternative interaction methods such as their mouths - a vital organ in their everyday life for making art (e.g. painting) and in locomotion (e.g. operating wheelchairs).

This case study aims to address a similar concern by examining the development of the XR artwork *Crip Sensorama*, co-created by a non-disabled HCI researcher and disabled artists Eric and Christian.

Concretely, through this study we aim to share how our inter-able¹ team adopted the methodology of criptastic hacking [31] in the design and development of the artwork. Criptastic hacking as a methodology embraces the lived experiences of disability and the adaptive hacking and tinkering often employed by people with disabilities. This method led us to develop accessible mouth gestures for Eric and Christian to navigate and interact in XR using a camera-based interface, designed to recognize specific mouth gestures. Later, these gestures were incorporated into *Crip Sensorama* (in two versions), based on the preferences and unique needs of each artist.

Overall, reflecting on the challenges faced in the design and development of the mouth interface for the artwork, we aim to share valuable insights to the HCI community who wish to create accessible XR solutions in future for people with severe motor disabilities. The paper is structured as follows: Section 2 provides the context of working with the two artists and significance of using criptastic hacking as a methodology, leading to creation of mouth gestures as an alternative input modality in XR. In Section 3 we discuss the design decisions, choice of gestures and the developed prototypes. In Section 4, we describe the two versions of the artwork, *Crip Sensorama*, tailored to the needs of the two artists. In Section 5 we discuss our insights through a detailed reflection of the design process. Sections 6 and 7 discuss the current limitations and the future work respectively, followed by a conclusion in Section 8.

2 Context and Significance

To give a brief context, Eric Desrosiers and Christian Bayerlein are two disabled artists and activists living in Montreal, Canada and Koblenz, Germany, respectively. The following section describes their disabilities and their artistic practices in order to highlight how *Crip Sensorama* as an artwork draws on their unique experiences and the innovative hacks employed by them in their everyday living.

2.1 The disabled artists - Eric and Christian

Eric Desrosiers is a painter in his mid 50's who has muscular dystrophy, a degenerative neuromuscular disease. He does not have a formal education in art but has self-learned painting with watercolor, oil, and acrylic. As he mentions in his CV, "*with my physical and pulmonary limitations I discovered a talent for music and song*" but "*in the winter when the weather does not allow me to perform, I like to turn to painting*". However, not able to paint using his hands, he hacks and tinkers with his home environment to find plausible technical solutions to paint. For instance, Eric uses a mouth-controlled joystick called Jouse3 to operate a robotic arm by Kinova [12], precisely controlling robotic arm's movements through pushing, rotating, blowing, and sucking on the joystick with his mouth. Furthermore, attaching a spherical rubber ball to the finger-like-jaws of the arm (see Figure 1) which further holds and manipulates his paintbrush, Eric along with his caretakers has carefully configured this customized setup to continue his passion to paint. By moving his chin, jaw, and lips, he thus moves the robotic arm to dip the

brush into paint and apply strokes to the canvas. Through this intricate combination of mouth-controlled gestures and creative-hacks, Eric has developed a unique dexterity in mouth movements to express his artistic creativity - one which he wanted to re-create in the form of an XR experience.



Figure 1: Eric using his mouth-controlled joystick to move the robotic arm to paint

In contrast, Christian Bayerlein is an artist, technologist, and activist who has Spinal Muscular Atrophy (SMA) due to which he has almost no movements in the upper and lower parts of his body. In his late 40's, Christian is a hacker and maker and he uses technology as a tool for activism around disability, as he writes on his website "*I fight for the rights of disabled people, e.g. for better accessibility or the right to live in the community*". Christian loves to travel around the world in his wheelchair which is driven using an integrated self-hacked Mo-vis joystick. He has modified the joystick by extending it using a straw, enabling him to control it entirely with his mouth. While Christian has tried VR headsets with eye-tracking and brain-controlled interfaces (BCI) [25], when we first met, he questioned: "Can I use my mouth to navigate in VR?". This question highlighted his struggle with bulky headsets which have eye-tracking as a feature but are difficult for users like Christian who have limited motor control in their necks. Christian, cannot easily keep his head upright when wearing such heavy devices, hence, he wished to use mouth as an input modality in XR, integrated to a lightweight headset like Quest 2 or Quest 3. Like Eric, Christian wanted to create an XR experience through which he could enable the non-disabled audience, experience and encounter his lived experiences of disability (such as drinking coffee using a straw) through an interaction based XR artwork.

2.2 Criptastic Hacking

Eric and Christian's lived experiences of disability encourage them to hack and tinker with their everyday environment, tools, and technologies. This is what disability scholar Remi M. Yergeau calls 'criptastic hacking' [31]. It's a "methodological approach that harnesses and draws on the long history of hacking and tinkering performed by disabled people to navigate in an inaccessible world" [23]. Following the same approach, embracing the experiences of Eric and Christian to hack technologies to suit their abilities (e.g. robotic arms to paint using mouth), first, this case study shares the

¹The term inter-able is commonly used to describe romantic relationships between a disabled and non-disabled individual. We re-frame it here to refer to a team of individuals with diverse disabilities and non-disabled folks.

design and development of a mouth-controlled interface adapted and customized for Eric and Christian. Second, we demonstrate how inter-able collaborations (disabled artists and the non-disabled HCI researcher) to create an XR artwork challenged the normative assumptions in HCI (e.g., reliance on hand-operated controllers in XR). Finally, through the description of the two versions of artwork, *Crip Sensorama*, we illustrate how integrating mouth-controlled gestures in XR enable us to not only to adapt to the specific needs of disabled artists but develop a novel and unconventional interaction modality in XR.

2.3 Crip Sensorama

Based on the unique needs of Eric and Christian, this case study presents the two versions of the developed XR artwork, *Crip Sensorama*, which we will refer to as CS1.0 and CS2.0. The title 'Crip Sensorama' is a humble nod to the physical device 'The Sensorama' (often framed as the earliest VR technology) built in 1962 by filmmaker Morton Heilig [7]. As a machine, Sensorama invited the audience to immerse themselves in multisensory stimuli such as stereoscopic colour display, fans, odor-emitters, stereo-sound system, and a motional chair. However, the device, a spectacle of that time which *simulated immersive environments such as riding a motorcycle through New York* was clearly not accessible for disabled bodies. Crip Sensorama therefore re-imagines this historical much cited example in XR to address the current accessibility problems in XR by using the reclamation of the word 'Crip' from "Cripple" as a marker of proud and defiant identification of disabled bodies [18]. Overall, CS1.0 and CS2.0 are XR artworks experienced using Meta Quest 3 headsets integrated with a customized and adapted camera-based mouth tracking device (from HTC VIVE but now discontinued) co-developed in collaboration with Eric and Christian. We now describe the design and development of the mouth interface prototypes in Section 3.

3 Design Process

To create the mouth-gesture interface for Eric and Christian, we employed an iterative design approach, informed by the principles of criptastic hacking. Criptastic hacking is a form of DIY (do-it-yourself), grassroots problem-solving where disabled individuals creatively adapt, hack, or reconfigure technologies to fit their unique needs and experiences. The combination of cript-hacking with an iterative design methodology was crucial as it allowed us to refine the design through multiple prototyping phases, where Eric and Christian contributed not only as users but as active co-designers. Their role as hackers, shaped by their lived experiences of disability, enabled them to influence and shape the prototyping process, ensuring that the interface was customized to their specific needs and abilities. The section below describes the development of two prototypes of the camera-based mouth-controlled interface in XR.

3.1 Prototypes for mouth interface

3.1.1 DIY RGB camera-based prototype. As a starting point, to detect the contours of the mouth, our first prototype used open-source face-mesh-landmark detection algorithm provided by Google-media-pipe [15] using a 1920 x 1080 resolution Lenovo web-cam.

This particular Face-mesh algorithm can detect 478 3-dimensional coordinates on the face such as points on the upper and lower lip, around the nose, eye-brows and eyes which are also referred to as the 'markers'. Since Eric and Christian cannot turn their heads (due to their disability) to achieve a 360-degree POV in VR, we designed three mouth-based gestures to facilitate rotation (clockwise and counterclockwise) in the VR environment. These gestures were based on the distances between specific 3D facial markers, detecting three features: 1) Opening mouth, 2) Moving lips laterally to the left, and 3) Moving lips laterally to the right. For example, the distance between the central markers on the upper and lower lips was used to detect when the mouth was open. When this distance exceeded a threshold, it triggered clockwise rotation, allowing the artists to rotate the VR scene (e.g. 15 degrees) by opening their mouths. Similarly, the distance of markers on the leftmost point on the lip and the rightmost point were calculated to detect lateral lip movements. This would in-result trigger counterclockwise rotation in VR for lateral movements of the lips towards left and likewise for right movement (triggering clockwise rotation).

However, the face-mesh algorithm needs a full view of the face to detect all its markers. Since wearing a VR headset hides the eyes and upper part of the nose, face-mesh algorithm could not identify the lower part of the face, and thus, the mouth. An approach suggested by the HCI researcher (in the team) to solve this problem was to develop or find an existing face-mesh algorithm that works with occlusion of human faces. However, with Eric and Christian's active participation in the prototyping, the team found another workable hack.



Figure 2: Hacking existing face-mesh detection algorithms by pasting paper masks of faces of famous celebrities on Meta Quest headsets

Instead of finding another machine-learning algorithm, the team decided to rather confuse the existing algorithm to make it work. We printed a few paper masks of the faces of famous celebrities (e.g. Angelina Jolie's) and pasted them on the Meta Quest headset such that the eyes and upper part of Jolie's nose juxtaposes with the lower part of the mouth of the artists (as shown in Figure 2). This hack, although not perfect, tracked the occluded face and hence

“the mouth”, enabling the artists to rotate the virtual camera in VR and have a 360 degree point of view through mouth movements. However, this hack had its limitations, as the camera needed to remain stationary to keep tracking the artists’ faces, restricting their ability to move their wheelchairs freely, which would cause them to slip out of the camera’s view. Thereby, the team moved onto another solution where the mouth could be tracked invariant of the movement of the XR headset.

3.1.2 Infrared camera-based prototype. During the process of building the DIY solution, the team discovered that HTC Vive, a leading seller of XR headsets, had been also selling a facial tracker (now discontinued) that allowed users to replicate their lip and mouth movements in VR chatrooms such as VRChat [28]. This feature is beneficial for users joining social-media platforms in XR where their mouth movements are mirrored onto their avatar’s face while communicating with others. Upon purchasing the tracker, the team utilized the provided Software Development Kit (SDK) and discovered that the internal AI algorithm classified 37 mouth gestures to facilitate precise replication of mouth movements in XR. The team decided to adapt this tracker for use with Meta Quest 2 and later with Meta Quest 3 headsets. It is important to note that HTC Vive only provides instructions for integrating the tracker with their own XR headsets. Thus, the team had to find ways to configure it with the Quest headsets by creating customized 3-D printed mounts for the tracker to mount onto the two headsets (as shown in Figures 3 and 4). Finally, based on specific needs of the artists, we selected a set of gestures from the list of described gestures in SDK (please refer to Appendix A), to enable movement and interaction in XR. The next section describes the choice of gestures for desired movement and interaction in XR by the artists.

3.2 Choice of mouth gestures for desired navigation and interaction

The SDK of the HTC Vive facial tracker provides a PDF explaining the classified 37 gestures through image representation of an avatar (See Appendix A for complete list of gestures). Eric selected a set of seven mouth gestures from the PDF list: 1) Jaw_Open (opening the mouth), 2) Tongue_LongStep2 (extending the tongue to the longest), 3) Cheek_Puff_Left (puffing the left side of the cheek), 4) Cheek_Puff_Right (puffing the right side of the cheek), 5) Mouth_Lower_Right (moving the lower lip further rightward), 6) Mouth_Lower_Left (moving the lower lip further leftward) and 7) Mouth_Pout (pouting or moving lips forward). After testing the gestures in VR (through a UNITY scene representing the software that detected the gestures), Eric finalized six gestures to move and navigate in VR. Since internally, each gesture was classified with a probabilistic value (per frame or per update in UNITY) ranging in-between 0 and 1, we used a threshold for triggering the desired movement in VR. For example, if the probability value for Tongue_LongStep2 gesture was above the choosen threshold of 0.6, the virtual camera will start moving forward, that is, in the direction where the VR headset is facing. These values were selected by testing the gestures in the software represented inside the headset, customizing it for both, Eric and Christian.

The list of selected gestures for Eric and corresponding movements in VR are shown in Table 1 which were later integrated in



(a) Eric, moving his tongue out to move forward in VR



(b) Christian, opening his mouth to control passthrough opacity in XR

Figure 3: Eric and Christian using mouth gestures to navigate and interact in XR using the mouth interface.

CS1.0. The same list of gestures were proposed to Christian. While Christian preferred the choice of gestures by Eric, he selected four gestures from the list to navigate and move in VR (as shown in Table 2). In addition, he decided to use them for interaction as well such as control passthrough opacity, that is the transparency of the video-feed inside the headset while using the passthrough feature of Quest headset. This interaction modality was later adapted in CS2.0 to produce an aesthetic effect. Interestingly, while Eric and Christian share similar physical limitations, their gesture preferences exhibited personal variation despite significant overlap. This factor illustrates how the flexibility inherent in criptastic hacking enabled us to embrace the adaptability of individual disabled bodies while simultaneously leveraging the shared affordances that emerge through collaborative, user-driven hacking processes. The overall gestures selected for Eric and Christian to move, navigate and interact in XR are described in Table 1 and 2.

4 Art works

4.0.1 CS1.0. The first version of the artwork was developed with (and for) Eric Desrosiers. Eric’s disability does not allow him to travel outside his hometown Montreal, Canada. However, Eric wanted to expand the visibility of his mouth-painted paintings to a broader audience. For CS1.0, which was exhibited as a part of gallery exhibition at xCoAx Conference in Treviso, Italy in July,

Gestures for Eric	Movement or Interaction
Tongue_LongStep2	Move Forward
Mouth_Pout	Move Backward
Cheek_Puff_Left	Rotate Counter-clockwise
Cheek_Puff_Right	Rotate Clockwise
Mouth_Lower_Right	Move right
Mouth_Lower_Left	Move left

Table 1: Choice of gestures for Eric to move and navigate in XR

Gestures for Christian	Movement or Interaction
Jaw_Open	Control passthrough opacity
Tongue_LongStep2	Move Forward
Cheek_Puff_Left	Rotate Counter-clockwise
Cheek_Puff_Right	Rotate Clockwise

Table 2: Choice of gestures for Christian to move, navigate and interact in XR

2024 [9], firstly, we developed a set of six mouth gestures (e.g. moving tongue out, puffing cheeks) as a means of navigation in VR for Eric. For instance, a pout gesture would enable Eric to move backward in VR while moving the tongue out, would enable him to move forward. As a part of the VR experience, we developed a 3D art gallery where digitalized versions of Eric's paintings (his five favourite ones) were placed in virtual rooms. In order to view his paintings, the audience (one at a time) were instructed inside VR to navigate in this VR gallery by using the mouth gestures developed and mapped onto Eric's mouth movements (See Figure 4a). The act of using one's mouth to navigate in VR space thus allowed the audience to pay attention to their mouth movements which are often overlooked but are crucial for Eric in his artistic practices, while, also revealing to the general public - an uncommon means of interaction in XR through art.

4.0.2 CS2.0. The second version of *Crip Sensorama*, CS2.0, was exhibited at Ars Electronica Festival for art, technology, and society in Linz, Austria in September, 2024 [10]. CS2.0 was developed with (and for) Christian where the audience could experience the ordinary acts in the everyday life of Christian such as drinking a cup of coffee. For context, Christian begins his day by lying on bed or in his wheelchair with a coffee. Since he doesn't require a handle to hold the cup, his coffee cup is custom made without handle and comes with a straw. Christian's partner/caretaker prepares the coffee and puts it beside him, which he drinks and sips slowly (sometimes for more than 30 minutes) using a straw. However, Christian did not want to re-enact this act in XR but instead use it as an artistic device to carry forward a narrative around his disability in XR. To accomplish this, first, we tested the developed gestures for Eric on Christian, narrowing them down to a customized set of 4 mouth gestures (as per Christian preferences) for navigation in VR. In addition, we also utilized the same gestures to control the passthrough opacity (see-through feature in Quest 3 which allows the shift from virtual to mixed-reality) and activate audio sources (with narration from Christian) in CS2.0. As a part of the XR experience in CS2.0,



(a) A visitor moving their tongue out to move in VR Gallery, viewing Eric's paintings in CS1.0



(b) A visitor interacting in mixed reality by opening her mouth, holding Christian's straw, in CS2.0

Figure 4: Visitors in CS1.0 and CS2.0, navigating and interacting in XR using their mouth gestures

an audience of 2 visitors, experiences a narrative around Christian's life staged in three acts: 1) A VR scene in which the audience navigates and move within a virtual mouth using their mouth gestures (previously mapped and parameterized on Christian), 2) A mixed reality scene enabled through the passthrough technology where audience use mouth gestures to activate Christian's narration about his experience of living with disability, and 3) A mixed-reality scene where the audience control the passthrough opacity through mouth movements to shift from virtual to mixed reality, enabling them to interact with the elements present in the physical space (e.g. to hold Christian physical cup or straws as shown in Figure 4b).

5 Reflection

This section provides the insights from our reflection of adopting the methodology of criptastic hacking in co-designing with the disabled artists. We thus share three themes which emerged during our design process and can, henceforth, encourage inclusive accessible XR design : 1) Lived Experience(s) for Design, 2) Adapting Prototypes by Embracing Users Creativity, 3) Inter-able Co-Design for Disability Empowerment and Inclusion.

5.1 Lived Experience(s) for Design

The common means of interaction in the current generation of XR technologies is through hand-held controllers or free-hand gestures [8]. In the initial stages of the design thinking phase with the artists, the HCI researcher (non-disabled) on the team recommended using the micro-movements of Eric and Christian's fingers as a means of

navigation in VR. However, drawing on their everyday experiences with assistive technologies, the artists preferred using their mouths over their hands. This highlights how traditional approaches [2], where designers building on their biases towards the users, can reinforce ableism in the design process. As disability scholar Ashley Shew notes, such traditional approaches in designing for inclusion aim for “*empowering disabled people through technologies while at the same time reinforcing ableist tropes*” [23]. Criptastic hacking, on the other hand, embraces the lived experiences of disability, hence allowing us to create a more inclusive and effective design by creating two sets of mouth gestures to move and interact in XR, tailored to the needs of individual artists.

Another consideration during the design process was the potential use of voice-enabled interaction, where the disabled artists could use voice commands to navigate or interact in XR. However, the team decided against pursuing this approach for two key reasons. First, Christian’s speech is significantly affected by his disability that most AI-based voice recognition algorithms fail to interpret accurately. Additionally, the target setting for showcasing these XR interactions, such as art exhibitions like Ars Electronica, typically involves noisy environments that are not conducive to reliable voice recognition. Second, Eric and Christian rely heavily on physical mouth movements in their daily lives, such as using chin and lip motions to operate assistive devices like the Jouse3 (for Eric) or a wheelchair (for Christian). Hence, this existing reliance made voice commands less practical compared to leveraging mouth gestures for XR interaction.

Although both artists shared similar physical disabilities, their preferences for interaction modalities in XR were influenced by their distinct artistic practices and visions. Eric favored a set of six gestures that allowed him to achieve a full 360-degree view, including the ability to move backward in the virtual environment. In contrast, Christian opted against the backward movement gesture; instead preferring a two-step approach, which involved rotating first and then moving forward to reach the desired viewpoint. This divergence illustrates how individual artistic intentions can shape preferences for interaction, even among users with comparable disabilities.

Drawing on lived experiences of users, as exemplified by Eric and Christian, reveals how the design process can move beyond traditional, ableist perspectives. It highlights the value of learning from disability culture and practices. By centering the embodied experiences of disabled individuals, our chosen methodology demonstrates how preferences for interaction shaped by personal, creative, and functional needs can lead to unconventional solutions such as using the mouth as an interaction modality in XR.

5.2 Adapting Prototypes by Embracing Users Creativity

Criptastic hacking encourages an iterative and improvisational design approach, emphasizing flexibility and creativity from the users themselves. This process allows for user-driven adaptation, where technology can be reshaped to suit specific needs. In our case, the team was able to creatively re-purpose existing machine-learning algorithms, such as the face-mesh landmark detection system, which typically fails when facial features are occluded. By leveraging the

artists’ input and experimenting with unconventional solutions, we discovered an imperfect yet functional hack—confusing the algorithm by using Angelina Jolie face masks on the Meta Quest headsets. This inventive workaround enabled the facial recognition system to track the necessary mouth gestures, demonstrating how user-driven creativity can transform and adapt technologies that would otherwise be exclusionary. This experience highlights that in designing solutions for disabled communities (specifically when the designers are non-disabled), embracing the hacks and tinkering skills of the disabled users can offer more inclusivity.

5.3 Inter-able Co-Design for Disability Empowerment and Inclusion

During our design process with the manufactured HTC Vive facial tracker, we encountered a critical challenge with the AI’s detection capabilities, particularly concerning Christian’s use of the Mouth_Pout gesture. While this gesture functioned effectively for Eric, Christian’s malocclusion—a misalignment of teeth—interfered with the algorithm’s ability to accurately recognize it. Instead of detecting the intended Mouth_Pout, the system generated a false positive for Jaw_Open (i.e. mouth-open). This experience highlights how even minor anatomical differences can lead to significant disparities in how assistive technologies function, revealing biases in facial tracking algorithms that are often designed with a limited range of users in mind.

Our collaboration among individuals with diverse yet overlapping disabilities underscored the importance of inclusive design practices. Christian’s experience, shaped by his unique body-mind, brought to light biases that may have otherwise gone unnoticed in a more homogeneous design environment. Engaging in this inter-able co-design process allowed us to iteratively adapt our prototypes, refining the system based on the lived experiences of both artists. This adaptability became essential in the creation of the XR artworks Crip Sensorama (CS1.0 and CS2.0), encouraging creative problem-solving that addressed individual needs through a deeper understanding of disability’s complexities. Ultimately, the diverse perspectives and shared problem-solving of this inter-able team empowered both Eric and Christian, leading to more inclusive and innovative design solutions in XR technologies for creative expression.

6 Limitations

Since our focus was on creating immersive XR artworks solely with (and for) the two disabled artists, we did not conduct a formal gesture elicitation study [30] [27] for identifying a common set of mouth gestures to navigate and interact in XR. Moreover, the project was artist-centric, focusing on building XR experiences around Eric’s paintings and Christian’s act of drinking coffee. Therefore the team’s goal was limited catering to the aesthetics of the individual artworks and did not establish a universal applicable set of mouth gestures to navigate and interact in XR.

7 Future work

To enhance the understanding of mouth-based navigation and interaction in XR, we plan to conduct a formal gesture elicitation study aimed at identifying common (or multiple sets of) gestures

that can be utilized across a broader spectrum of users. As a part of our next study, we aim to hire a diverse group of participants with motor disabilities. This will enable us to gain insights into the variety of mouth gestures that may facilitate intuitive navigation in XR environments for individuals with severe disabilities. Currently, we employ a single mouth gesture (e.g., extending the tongue out) as input for movement or interaction in XR. In future work, we aim to explore using combinations of gestures for more complex movements. For example, while extending the tongue out could continue to move the user forward, opening the mouth midway could facilitate jumping while in motion within VR.

8 Conclusion

In this work, we illustrate that current XR technologies remain largely inaccessible to individuals with severe motor disabilities, but through the application of Cripstastic Hacking, we demonstrate a pathway toward more inclusive design. By collaborating closely with two disabled artists, we highlight how the creation of XR artworks — Crip Sensorama versions 1.0 and 2.0 — adopts unconventional interaction modalities, such as mouth gestures to enable people with severe motor disabilities to navigate, move, and interact in XR.

By reflecting on these experiences, we contribute valuable insights into how gesture-based navigation in XR can be adapted to meet the specific needs of disabled users such as using the mouth as an input-modality. Our work emphasizes the importance of integrating disabled users into every phase of the design process, allowing their creativity and expertise to drive technological innovation. We hope that this research will inspire future efforts in HCI and accessible XR design, encouraging researchers and practitioners alike to explore more equitable and adaptive design methods.

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A List of mouth gestures classified by the HTC Vive Facial Tracker

We here share the list of classified mouth gestures by the HTC Vive face tracker’s internal AI algorithm. HTC Vive provides two versions for lip and mouth tracker, v1 and v2. We chose the version v2 as it was the latest one at that time. Please refer to Figures 5 and 6 for the total list of 37 gestures.

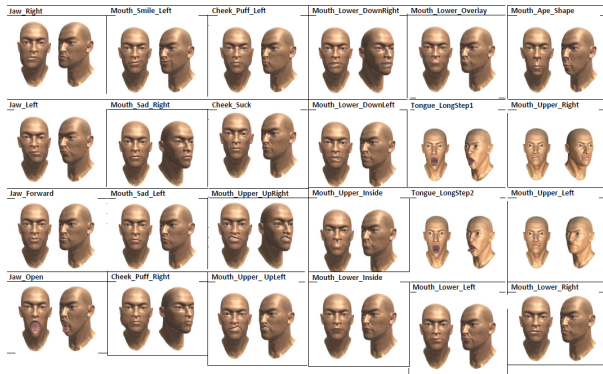


Figure 5: Mouth gestures (24) classified internally by HTC Vive faial tracker’s AI algorithm

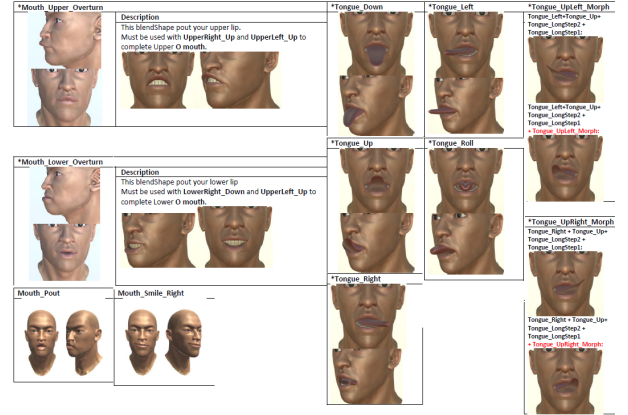


Figure 6: Mouth gestures (13) classified internally by HTC Vive faial tracker’s AI algorithm